Modeling reading by interpolating predictions from *n*-grams & lossy sentence memories

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Language model (LM) surprisal predicts human reading behavior [1, 2, 3, *inter alia*], though larger LMs show poorer fit to reading times (RTs) [4]. One explanation for this mismatch is that larger LMs have access to long context windows which humans cannot keep in memory [5, 6]. Recent work on lossy-context surprisal [7] has found that noising LM memories improves fits to RTs for multiple constructions [8, 9]. Most lossy-context models use *token-level* noise, where prior tokens are probabilistically retained based on their distance from the current input. However, it is unclear how this approach scales to multi-sentence reading: while people may not have exact memory of individual words in prior sentences, they do retain dense, abstract representations of previous material [10, 11]. **Model.** To capture lossy-context effects across sentences, we built a language model with three components: 1) an *n*-gram model, 2) an encoding layer, 3) and a hybrid memory system consisting of *r* recent sentences (*r* =10) and *o* older sentences (*o*=30). First, we compute 5-gram statistics of the training data (WikiText-2). Second, we use GPT2 to encode tokens for each word within a sentence, then, at sentence boundaries, compress these token-level vectors into a dense sentence vector. These dense vectors enter *r* with activation values that decay over time, simulating classic memory decay effects [7, 8, 9]. Following prior work on how unique sentence meanings are retained over time [13, 14], we push the least similar memory in *r* (using standardized cosine similarity) to *o* when *r* is full. During language modeling, we mean-pool the GPT2 vectors of the *n*-gram context & calculate its similarity to dense memories in both *r* and *o*. So the model can use lossy sentence memories to inform language modeling, we estimate the most useful memories using retrieval weights as the product of these similarities and the memories' activation strengths. We project the retrieved representations from *r* and *o* through separate linear layers to produce logits over the vocabulary, and then average the logits together. Finally, the model interpolates probabilities from the *n*-gram and memory components based on a fixed parameter (α), allowing the model to balance local statistical patterns with information from prior sentences. We tested five values of α {0 (only memory), 0.2, 0.4, 0.6, 0.8, 1 (only *n*-grams)}. **Experiment**. To determine how well our model aligns with multi-sentence reading in English, we collected per-word surprisal estimates for the Natural Stories Corpus [15] to predict by-word reading times (RTs). We calculated the difference in log-likelihood (LL) between a baseline linear mixed-effects model and a model that includes surprisal. As seen in Fig. 1, the best fitting model uses surprisal calculated *mostly* from *n*-gram statistics (α=0.8); the worst fit uses surprisal calculated *only* from *n*-gram statistics (α=1). **The trade-off between *n*-gram & memory predictions**. To probe why the two most predictive models (α=0.2, α=0.8) performed best, we computed the difference between the squared residual errors for each word *w* [$res_{0.8}(w)$- $res_{0.2}(w)$], adapting [6]: values > 0 indicate that relying on memory (α=0.2) was more predictive for that word. First, we measured r^2 differences by dependency lengths (ℓ) for prior words (i.e. only dependencies that refer back); we find short dependencies (ℓ<5) are most common in the data (Fig. 2), possibly explaining why the *n*-gram-reliant model is most predictive. Second, we split each story into 5 regions (1 = first fifth of story, etc.) and calculated r^2 differences; surprisingly, we find that the memory-reliant model is more predictive in earlier regions (Fig. 3). In sum, our best models rely on a blend of predictions from both *n*-grams & lossy sentence vectors (compared to either extreme), supporting a model of reading that balances both memory sources and accesses some (but not all) prior context.

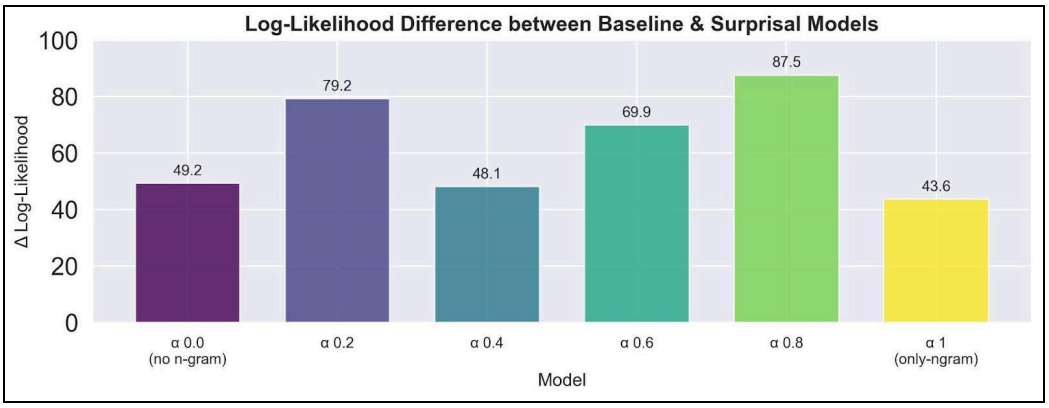

Figure 1: Log-likelihood differences between baseline & surprisal linear mixed-effects models. Lighter colors indicate greater α thresholds (more *n*-gram reliance).

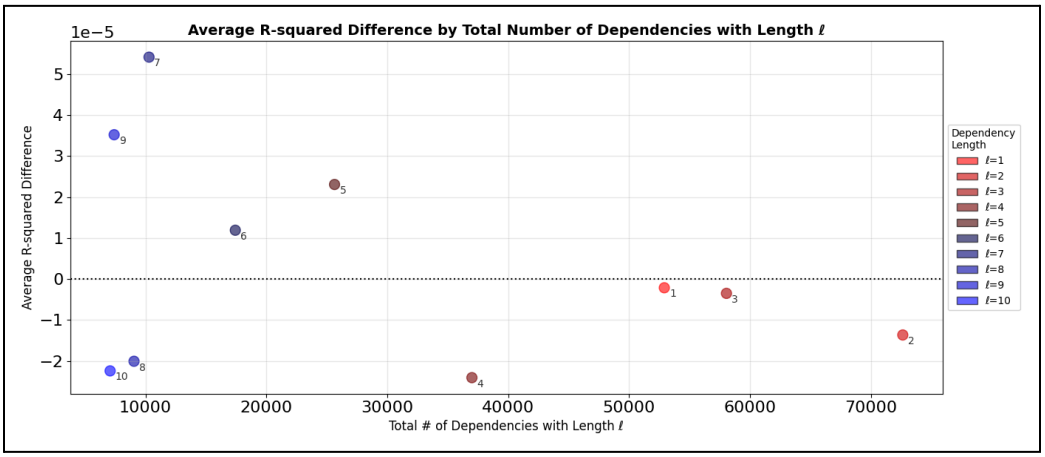

Figure 2: Dependency-length analysis of model residual differences.

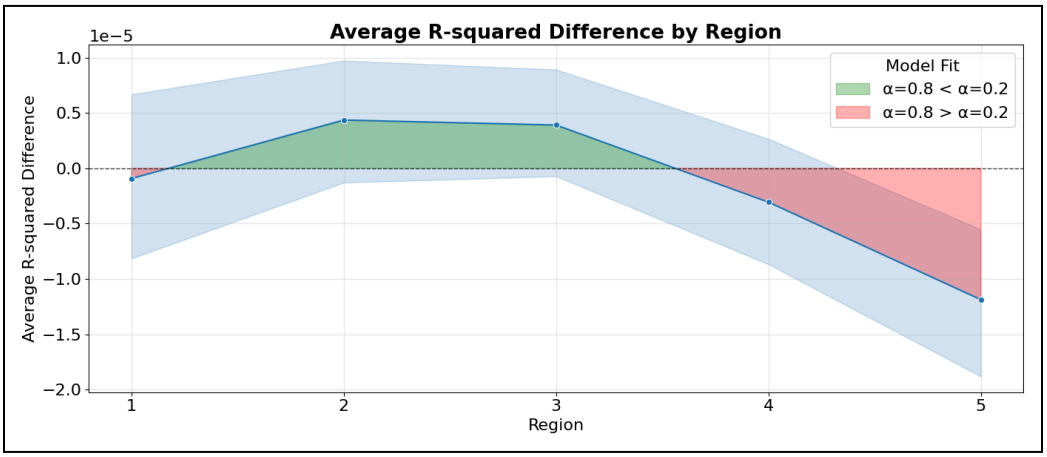

Figure 3: Region-based analysis of model residual differences..

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